

DSpark: Con dence-Scheduled Speculative Decoding with Semi-Autoregressive Generation

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Abstract

Speculative decoding accelerates Large Language Model (LLM) inference by decoupling draft generation from target verification. While recent parallel drafters efficiently propose long token sequences in a single forward pass, they suffer from rapid acceptance decay due to a lack of inter-token dependencies. Furthermore, indiscriminately verifying these extended blocks wastes critical batch capacity on tokens with high rejection risks, severely degrading throughput in high-concurrency serving systems. We introduce DSpark, a speculative decoding framework that unifies high-throughput parallel generation with adaptive, load-aware verification. To maintain draft quality, DSpark utilizes a semi-autoregressive architecture—coupling a parallel backbone with a lightweight sequential module—to introduce intra-block dependency modeling and mitigate suffix decay. To optimize system efficiency, DSpark employs confidence-scheduled verification, dynamically tailoring the verification length for each request based on estimated prefix survival probabilities and engine-specific throughput profiles. On of nine benchmarks across diverse domains, DSpark substantially improves the accepted length over state-of-the-art autoregressive and parallel drafters. When deployed within the DeepSeek-V4 serving system under live user traffic, DSpark successfully mitigates verification waste. Compared to the established production baseline (MTP-1), DSpark accelerates per-user generation speeds by 60%–85% at matched throughput levels. More importantly, by preventing severe throughput degradation under strict interactivity constraints, it enables performance tiers that were previously unattainable, shifting the Pareto frontier of our serving system. To facilitate community progress, we open-source the [DSpark checkpoints](#) alongside [DeepSpec](#), an algorithm-driven training repository for speculative decoding.

1. Introduction

Large Language Models (LLMs) generate text autoregressively: each new token requires a full forward pass conditioned on all preceding tokens, making inference latency proportional to the output length. The resulting low GPU utilization and high user-perceived waiting time constitute

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